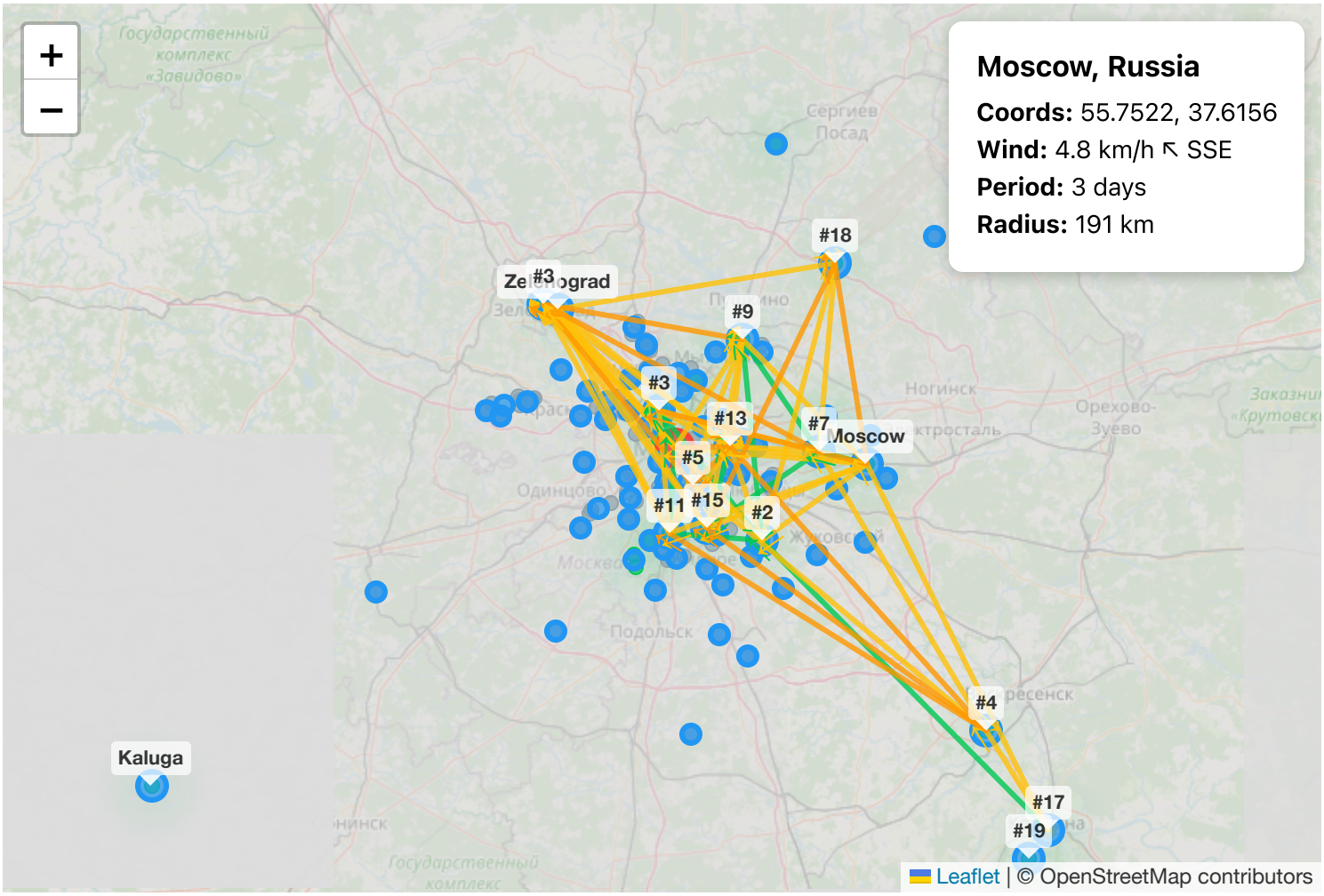


# Air Quality Report — Moscow, Russia

## Key Insights

- **Highest PM2.5:** 179.3 µg/m³ (AQI 230, Very Unhealthy) at Moscow-5 — 2026-03-15 04:00
- **Strongest front:** Moscow-15 → Moscow-13 at 16 km/h N (correlation 94%, lag 1h)
- **Wind:** 4.8 km/h SSE — calm — pollutants tend to accumulate locally
- **Assessment:** Very unhealthy to hazardous. Avoid outdoor activity. Close windows.

## Spikes

| City | Time | PM2.5 | Delta | Z-score |
|---|---|---|---|---|
| Moscow, Russia | 2026-03-15 15:00 | 5.8 | +2.1 | 2.1 |
| Zelenograd | 2026-03-15 18:00 | 6.4 | +1.8 | 2.6 |
| Vladimir | 2026-03-14 02:00 | 8.3 | +3.0 | 2.4 |
| Vladimir | 2026-03-14 15:00 | 6.8 | +2.6 | 2.0 |

| City | Time | PM2.5 | Delta | Z-score |
|---|---|---|---|---|
| Vladimir | 2026-03-14 20:00 | 8.8 | +3.4 | 2.7 |
| Moscow | 2026-03-14 03:00 | 9.0 | +2.5 | 2.3 |
| Moscow | 2026-03-15 22:00 | 9.2 | +2.5 | 2.3 |
| Moscow-2 | 2026-03-15 05:00 | 9.6 | +3.4 | 2.7 |
| Moscow-3 (2 sensors) | 2026-03-14 02:00 | 6.4 | +2.3 | 2.8 |
| Moscow-4 | 2026-03-15 19:00 | 12.2 | +6.8 | 4.0 |
| Moscow-5 | 2026-03-15 04:00 | 179.3 | +174.3 | 5.9 |
| Moscow-7 (3 sensors) | 2026-03-14 04:00 | 11.4 | +2.4 | 2.5 |
| Moscow-9 | 2026-03-14 06:00 | 10.1 | +2.3 | 2.1 |
| Moscow-9 | 2026-03-15 16:00 | 5.8 | +2.5 | 2.2 |
| Moscow-11 | 2026-03-14 02:00 | 6.6 | +1.6 | 2.0 |
| Kaluga | 2026-03-15 03:00 | 2.9 | +1.6 | 2.8 |
| Moscow-13 | 2026-03-14 05:00 | 8.7 | +1.7 | 2.4 |
| Moscow-13 | 2026-03-15 15:00 | 5.2 | +1.5 | 2.0 |
| Zelenograd-3 (2 sensors) | 2026-03-14 05:00 | 2.4 | +0.6 | 2.1 |
| Zelenograd-3 (2 sensors) | 2026-03-15 17:00 | 1.5 | +0.7 | 2.3 |
| Zelenograd-3 (2 sensors) | 2026-03-15 18:00 | 2.1 | +0.6 | 2.3 |
| Moscow-15 | 2026-03-14 03:00 | 16.9 | +4.0 | 2.0 |
| Moscow-17 | 2026-03-14 23:00 | 9.9 | +4.5 | 2.5 |
| Moscow-17 | 2026-03-15 10:00 | 13.7 | +3.8 | 2.1 |
| Moscow-18 | 2026-03-14 06:00 | 15.6 | +3.9 | 2.1 |
| Moscow-18 | 2026-03-14 17:00 | 14.8 | +7.7 | 4.0 |
| Moscow-19 | 2026-03-15 11:00 | 49.6 | +45.1 | 4.8 |

## Pollution Fronts

| From → To | Speed (km/h) | Direction | Lag (h) | Correlation |
|---|---|---|---|---|
| Moscow-15 → Moscow-13 | 16 | N | 1 | 94% |
| Moscow → Moscow-7 (3 sensors) | 10 | W | 1 | 91% |
| Moscow-7 (3 sensors) → Moscow-15 | 26 | SW | 1 | 91% |
| Moscow, Russia → Moscow-3 (2 sensors) | 8 | NW | 1 | 91% |
| Moscow-17 → Moscow-2 | 80 | NW | 1 | 89% |
| Moscow-2 → Moscow-3 (2 sensors) | 32 | NW | 1 | 89% |
| Moscow-2 → Moscow-9 | 39 | N | 1 | 89% |
| Moscow-11 → Moscow-2 | 18 | E | 1 | 88% |
| Moscow-11 → Moscow, Russia | 16 | N | 1 | 88% |
| Moscow-7 (3 sensors) → Moscow-9 | 26 | NW | 1 | 86% |
| Moscow-2 → Moscow-13 | 19 | N | 1 | 84% |
| Moscow-7 (3 sensors) → Moscow-13 | 17 | W | 1 | 84% |
| Moscow, Russia → Zelenograd | 35 | NW | 1 | 84% |
| Moscow → Moscow-9 | 17 | NW | 2 | 83% |
| Moscow-11 → Moscow-9 | 20 | N | 2 | 83% |
| Moscow-13 → Moscow-9 | 21 | N | 1 | 81% |
| Moscow, Russia → Moscow-9 | 25 | NE | 1 | 80% |
| Moscow → Moscow-2 | 25 | SW | 1 | 80% |
| Moscow-11 → Moscow-3 (2 sensors) | 24 | N | 1 | 80% |
| Moscow-11 → Moscow-13 | 20 | NE | 1 | 78% |
| Zelenograd → Moscow-15 | 51 | SE | 1 | 78% |
| Zelenograd → Zelenograd-3 (2 sensors) | 3 | W | 1 | 78% |
| Moscow-13 → Zelenograd-3 (2 sensors) | 23 | NW | 2 | 78% |
| Moscow, Russia → Zelenograd-3 (2 sensors) | 19 | NW | 2 | 78% |
| Moscow → Moscow-15 | 17 | W | 2 | 77% |

| From → To | Speed (km/h) | Direction | Lag (h) | Correlation |
|---|---|---|---|---|
| Moscow-7 (3 sensors) → Moscow-3 (2 sensors) | 32 | W | 1 | 76% |
| Moscow-17 → Moscow-4 | 23 | NW | 1 | 76% |
| Moscow-15 → Moscow-9 | 37 | N | 1 | 76% |
| Moscow → Moscow-13 | 13 | W | 2 | 76% |
| Moscow-4 → Moscow-7 (3 sensors) | 63 | NW | 1 | 75% |
| Moscow-11 → Zelenograd | 48 | NW | 1 | 75% |
| Moscow-5 → Moscow-4 | 5 | SE | 15 | 75% |
| Zelenograd → Moscow-18 | 54 | E | 1 | 74% |
| Moscow-7 (3 sensors) → Moscow-18 | 36 | N | 1 | 74% |
| Moscow-11 → Zelenograd-3 (2 sensors) | 25 | NW | 2 | 74% |
| Moscow → Moscow, Russia | 37 | W | 1 | 74% |
| Moscow-2 → Zelenograd-3 (2 sensors) | 31 | NW | 2 | 74% |
| Moscow-2 → Moscow-18 | 55 | N | 1 | 74% |
| Moscow-17 → Moscow | 80 | NW | 1 | 73% |
| Moscow → Moscow-11 | 40 | W | 1 | 72% |
| Moscow-3 (2 sensors) → Zelenograd-3 (2 sensors) | 15 | NW | 2 | 71% |
| Moscow-15 → Zelenograd-3 (2 sensors) | 13 | NW | 4 | 71% |
| Moscow-3 (2 sensors) → Moscow-13 | 8 | SE | 2 | 71% |
| Zelenograd → Moscow-13 | 43 | SE | 1 | 71% |

## Methodology

**Data sources:** Open-Meteo atmospheric model (~11km grid) + Sensor.Community citizen science network (134 sensors, 86 analysis nodes).

**Spike detection:** Z-score on hourly PM2.5 differences. A spike is flagged when the rate of change exceeds 2 standard deviations above the rolling mean — indicating a sudden pollution event rather than normal variation.

**Front tracking:** Cross-correlation analysis between city pairs with time lags from -24h to +24h. The lag at peak correlation reveals how many hours pollution takes to travel between two points. Combined with haversine distance, this gives front speed and direction.

**Confidence:** When both Open-Meteo (model) and Sensor.Community (ground sensors) agree on lag direction, confidence is boosted. Disagreement reduces confidence, highlighting uncertain fronts.

## AQI Reference

| AQI | Category | Health Guidance |
|---|---|---|
| 0–50 | Good | No restrictions |
| 51–100 | Moderate | Sensitive individuals: limit prolonged outdoor exertion |
| 101–150 | Unhealthy for Sensitive | Active children and adults: reduce prolonged outdoor exertion |
| 151–200 | Unhealthy | Everyone: reduce outdoor exertion |
| 201–300 | Very Unhealthy | Everyone: avoid outdoor activity |
| 301–500 | Hazardous | Stay indoors. Close windows. |

Generated by airq — open source CLI air quality analyzer